Checkout 5xx during checkout-api 12.4.0 rollout

INC-5641 · **Closed** · opened by d.fischer on 2026-08-24 · labels: incident, sev::2, postmortem-done

Severity	SEV-2 (major)
Window	2026-08-24 14:07 – 14:26 CEST (19 minutes)
Resolved	2026-08-24 14:26 CEST
Affected systems	checkout-api, orders-db
Linked	PLAT-951, PLAT-952, PLAT-953

Description

During the rolling deploy of checkout-api 12.4.0, 8.2% of checkout requests failed with 502 or 504 for 19 minutes. Every failing request was routed to a newly started pod. Pods that had not yet been replaced served normally. We estimate CHF 14,900 of lost revenue from customers who did not retry.

Root cause

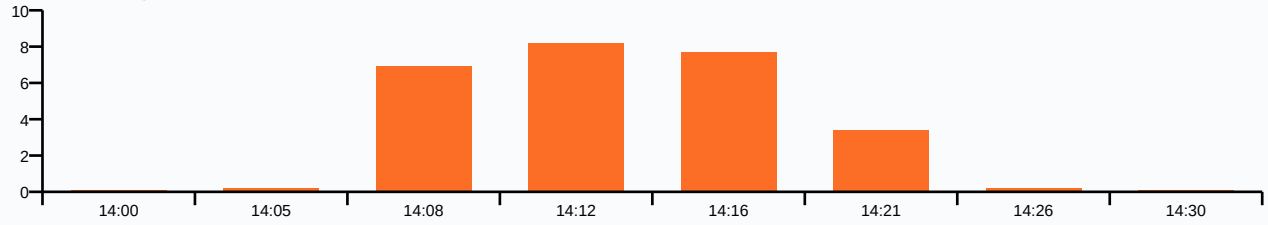
Two causes were jointly necessary; neither alone would have produced customer impact.

- **A startup migration held a table lock.** 12.4.0 added an index to `orders_idempotency` through a migration that runs on application startup, without `CONCURRENTLY`. The first new pod took an exclusive lock on the table for about four minutes, and every request needing that table blocked behind it, on every pod that tried to start while the lock was held.
- **The readiness probe did not check readiness.** The probe pointed at `/healthz`, which returns 200 as soon as the HTTP server is listening. New pods were marked ready and received traffic while still blocked on the migration. Had the probe pointed at `/ready`, those pods would have been held out of rotation and the old pods would have kept serving.

Timeline (CEST)

```
14:05  deploy of 12.4.0 starts (25% surge)
14:07  first new pod ready per /healthz; migration takes lock
14:09  CHECKOUT-5XX alert fires; d.fischer acknowledges
14:14  SEV-2 declared; deploy paused
14:19  pg_locks shows AccessExclusiveLock on orders_idempotency held by the migration
14:21  migration completes; lock released
14:26  new pods drain their queues; error rate at baseline; resolved
08-27  review held
```

checkout 5xx rate, %



Action items

Issue	Action	Owner	Due	State
PLAT-951	Remove schema migrations from application startup; run them as a separate, pre-deploy job with a lock timeout.	d.fischer	2026-09-11	Closed
PLAT-952	Point readiness probes at /ready on every service; fail CI when a readiness probe targets /healthz.	p.almeida	2026-09-04	Closed
PLAT-953	Migration linter: reject CREATE INDEX without CONCURRENTLY on tables above 1 GB.	k.oyelaran	2026-09-18	Closed

d.fischer closed this issue on 2026-09-01 · 9 comments hidden in export